

BALASUBRAMANI

Full Stack Developer | Backend Developer | SDE

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Objective

Full-stack engineer with 3 deployed production projects spanning PostgreSQL, MongoDB, and Firebase, and a 3-month Android internship shipping a real booking platform. Built and deployed JWT-authenticated REST APIs, configured GitHub Actions CI pipelines, and integrated ML microservices into Express backends. Looking for an SDE or Full-Stack role where backend depth and cross-stack ownership matter.

Skills

Languages: Python, JavaScript, SQL

Frontend: React.js, HTML5, CSS3

Backend & APIs: Node.js, Express.js, Django, REST API Design, JWT Authentication

Databases: PostgreSQL (relational schema, JOINS, transactions — hosted on Supabase), MongoDB (atomic operators), Firebase Firestore

Mobile: Android (Firebase Firestore, Geolocation API, Gemini AI integration)

DevOps & Tools: Git, GitHub, Postman, GitHub Actions (CI/CD), Vercel, Render, Netlify

Core CS: Data Structures & Algorithms, OOP Concepts, DBMS (normalized schemas, foreign keys, aggregate queries)

Experience

Android Developer Intern

February 2026 – May 2026

MindMatrix Technologies

Remote

- Built **Namma-Kelsa**, an Android booking app connecting daily-wage workers (painters, plumbers, electricians) with nearby customers — no middlemen. Used Firebase Firestore for real-time job listings and the Android Geolocation API for proximity-based worker discovery.
- Implemented GenAI-powered job matching using the Gemini API to compute description similarity between job posts and worker profiles; iterated across 3 mentor review cycles to refine matching accuracy and booking UX.

Projects

Job Tracker App | *React, Node.js, Express, PostgreSQL, JWT, GitHub Actions* | job-tracker-frontend-red.vercel.app 2026

- Designed a normalized PostgreSQL schema (users, jobs, applications tables) with foreign key constraints and relational queries; secured all endpoints with JWT + bcrypt authentication and parameterized inputs to prevent SQL injection.
- Deployed across a distributed cloud stack — React on Vercel, Node/Express on Render, PostgreSQL on Supabase — with environment-based dev/prod config; configured GitHub Actions CI to run ESLint and build checks on every push, keeping broken builds off production.
- Built input validation middleware across all REST endpoints, enforcing a consistent contract between frontend and backend and eliminating invalid-state errors in the API layer.

Book Barter System | *React, Node.js, Express.js, MongoDB, REST APIs* | bookbarterssystem.netlify.app 2025

- Debugged a concurrency issue in MongoDB write operations by restructuring queries to use atomic update operators (`$set`, `$inc`), eliminating data inconsistency under simultaneous user load — identified through manual load testing with Postman.
- Built and secured REST API endpoints for auth (JWT), book listings, and barter request workflows; applied middleware-level auth guards to protect all write operations from unauthenticated access.

BirthSense — Delivery Mode Predictor | *HTML, CSS, JavaScript, Node.js, Express.js, Python, Flask* | birthsense.netlify.app 2025

- Architected the backend integration layer: Express API received clinical input from the frontend, proxied it to a Python Flask microservice serving a Random Forest model (8,337 UCI patient records), and returned predictions — handling serialization and error states across two runtimes.
- Built the complete frontend (HTML/CSS/JS), defined the API contract between Express and Flask (input format, output schema, error structure), and managed end-to-end deployment across Netlify and Render with environment-based config.

Education

B.E. Computer Science Engineering

June 2026 (Expected)

HKBK College of Engineering

Bengaluru, Karnataka

- CGPA: **8.5 / 10**
- Relevant Coursework: Data Structures & Algorithms, OOP, DBMS & SQL, Operating Systems, Computer Networks, Software Engineering

Certifications

- NPTEL — Machine Learning | IIT Madras (via SWAYAM), 2024
- NPTEL — Cloud Computing | IIT Kharagpur (via SWAYAM), 2024